

RESEARCH & TECHNICAL NOTES — TRACE FINDERS

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SIH 2026 Problem Statement: SIH26189 — AI-Powered Criminal Network Analysis System

Subject Area: Graph Theory, Multi-Hop Evidence Fusion, Computer Vision ANPR Analytics, Explainable AI (XAI)

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1. Theoretical Graph Analytics & Centrality Models

Criminal network analysis relies on mathematical graph representations where entities are vertices V and evidentiary associations are directed or undirected weighted edges E .

$$G = (V, E, W)$$

1.1 Degree Centrality

Measures immediate connectivity and operational interaction volume:

$$C_D(v) = \frac{\deg(v)}{|V| - 1}$$

Where $\deg(v)$ represents the total number of distinct evidentiary links associated with node v .

1.2 Betweenness Centrality (Brokerage Detection)

Identifies critical "broker" nodes controlling communication or financial flow between disparate criminal syndicates:

$$C_B(v) = \sum_{s \neq v \neq t \in V} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Where σ_{st} is the total number of shortest paths from node s to node t and $\sigma_{st}(v)$ is the number of those paths that pass through node v . Nodes with high betweenness centrality are flagged as key logistical hubs or Hawala intermediaries.

1.3 PageRank Centrality (Syndicate Leader Identification)

Evaluates directional influence by propagating link weight recursively:

$$PR(p) = \frac{1-d}{N} + d \sum_{i \in M(p)} \frac{PR(i)}{L(i)}$$

Where $d = 0.85$ is the damping factor, $M(p)$ is the set of nodes linking to p , and $L(i)$ is the out-degree of node i .

2. Multi-Hop Cross-Domain Correlation Mechanics

Evidence in modern criminal investigations spans disparate data silos:

$\text{Domain}_1 (\text{Telecom}) \rightarrow \text{Domain}_2 (\text{ANPR CCTV}) \rightarrow \text{Domain}_3 (\text{Banking}) \rightarrow \text{Domain}_4 (\text{Blockchain})$

2.1 Spatio-Temporal Correlation Matrix

The correlation score $S_{\text{corr}}(A, B)$ between two events A and B is computed as:

$$S_{\text{corr}}(A, B) = w_t \cdot e^{-\lambda_t |t_A - t_B|} + w_s \cdot e^{-\lambda_s d(L_A, L_B)} + w_e \cdot \text{Sim}(E_A, E_B)$$

Where:

- t_A, t_B are event timestamps.
- $d(L_A, L_B)$ is the Haversine distance between geographical coordinates.
- $\text{Sim}(E_A, E_B)$ is entity attribute similarity (Jaro-Winkler distance on names/aliases).
- w_t, w_s, w_e are domain weighting constants (0.4, 0.35, 0.25).

3. CCTV / DVR ANPR & Facial Recognition Analytics

3.1 Automatic Number Plate Recognition (ANPR) Pipeline

- Frame Extraction & Preprocessing: Dual-pass contrast enhancement and CLAHE adaptive histogram equalization.
- Bounding Box Detection: YOLOv8-Nano vehicle license plate detector.
- OCR Feature Extraction: Tesseract-v5 & EasyOCR ensemble parser trained on Indian HSRP (High Security Registration Plate) font geometry.

3.2 24-Hour Timeline Marker Aggregation

Events from 12 active surveillance cameras (CAM-01 through CAM-12) are indexed into a temporal B-tree index, rendering real-time sighting markers across 24-hour windows (00:00 to 23:59 IST).

4. Explainable AI (XAI) & Legal Admissibility Framework

To satisfy judicial requirements under Section 65B of the Indian Evidence Act / Bharatiya Sakshya Adhiniyam, 2023, automated outputs must be fully transparent.

XAI ASSESSMENT STRUCTURE		
1. CORRELATION RATIONALE	→ Why these entities were linked	
2. FLAGGING REASONING	→ Suspicious pattern criteria	
3. CONFIDENCE METRIC	→ Calculated probability (e.g. 88%)	
4. ALTERNATIVE HYPOTHESIS	→ Defense counter-argument scenario	
5. EVIDENTIARY HASH REF	→ SHA-256 Chain of Custody Ref	

5. Entity Resolution & Deduplication Algorithms

When ambiguous candidates (e.g., Arjun S. vs Arjun Sharma) enter the system, the entity resolution engine evaluates identity likelihood using a multi-attribute similarity vector:

$$\mathbf{V}_{\text{match}} = \begin{bmatrix} \text{JaroWinkler}(\text{Name}_1, \text{Name}_2) \parallel \\ \text{GeographicOverlap}(\text{City}_1, \text{City}_2) \parallel \text{CommonCoContacts}(\text{CDR}_1, \text{CDR}_2) \end{bmatrix}$$

Match Confidence Score $M = \mathbf{w}^T \mathbf{V}_{\text{match}}$. Candidates scoring $0.40 \leq M < 0.75$ are placed in the Entity Resolution Review Queue for analyst verification.

6. References & Academic Literature

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